Arista Access Node MALL-ACC-80 Routing Protocol Flap Resolution SOP (Critical)

Node: MALL-ACC-80 | **Vendor:** Arista | **Severity:** Critical | **Alarm Code:** ALARM_3528

Description

The MALL-ACC-80 node located within the Malleswaram site is experiencing a Critical failure within its Access role due to a Routing Protocol Flap (BGP route churn, OSPF adjacency loss, ISIS LSP flooding) as indicated by the ALARM_3528 code. This failure is caused by the inherent complexities of the BGP, OSPF, and ISIS routing protocols which can lead to instability and flapping when faced with issues such as incorrect configuration, network topology changes, or hardware failures. The Arista EOS (Extensible Operating System) software, running on MALL-ACC-80, is a critical component in this failure domain as it handles the processing and propagation of routing information across the network. The root cause of this flapping is likely related to either a misconfiguration of the routing protocols, a hardware issue within the node, or a network-wide topology change that is causing the routing protocols to become unstable.

Pre-requisite Checks

- Verify the Malleswaram Node is accessible and the remote hands have been dispatched.

- Check the status of the redundant links connecting MALL-ACC-80 to ensure they are functioning properly.

- Ensure the Arista EOS software is up-to-date and running the latest version.

Potential Root Causes

Cause	Probability
Misconfigured routing protocols	High
Hardware failure within the node	Medium
Network-wide topology change	Low

Diagnostic Commands & Expected Outputs

```
Command: show bgp summary, Expected Output: BGP table version is 5, BGP table version is 5, Description: This command displays a summary of the BGP protocol, including the number of peers, routes, and the current BGP table version. This can help identify any issues with BGP connectivity or routing information.

Command: show ospf neighbor, Expected Output: Neighbor ID Pri State Dead Time Address Interface, Description: This command displays a list of OSPF neighbors, including their state and dead time. This can help identify any issues with OSPF adjacency or connectivity.
```

```
Command: show isis database, Expected Output: ISISv6 Database (LSP) - Link State, Description: This command displays the ISIS database, including the LSPs and their contents. This can help identify any issues with ISIS routing or topology.
```

Resolution Steps

Step 1: Step: Verify BGP routing information, Command: show bgp route , Description: This command displays the BGP routing information for the specified prefix. If the prefix is not being advertised or received, it may indicate a configuration issue or a hardware failure within the node.

Step 2: Step: Clear OSPF adjacency, Command: clear ospf process, Description: This command clears the OSPF process and restarts the adjacency. This can help resolve issues with OSPF connectivity or routing information.

Step 3: Step: Verify ISIS LSP flooding, Command: show isis database, Description: This command displays the ISIS database, including the LSPs and their contents. If the LSPs are not being flooded or received, it may indicate a configuration issue or a hardware failure within the node.

Post-Check Verification

```
Command: show bgp summary, Expected Output: BGP table version is 5, BGP table version is 5, Description: This command displays a summary of the BGP protocol, including the number of peers, routes, and the current BGP table version. This can help verify that the BGP protocol is stable and functioning properly.

Command: show ospf neighbor, Expected Output: Neighbor ID Pri State Dead Time Address Interface, Description: This command displays a list of OSPF neighbors, including their state and dead time. This can help verify that the OSPF protocol is stable and functioning properly.

Command: show isis database, Expected Output: ISISv6 Database (LSP) - Link State, Description: This command displays the ISIS database, including the LSPs and their contents. This can help verify that the ISIS protocol is stable and functioning properly.
```

Roll-back Procedures

- Step: Restore configuration, Command: copy run start, Description: This command restores the configuration to its previous state, rolling back any changes made during the resolution process.

- Step: Restart OSPF process, Command: clear ospf process, Description: This command restarts the OSPF process, rolling back any changes made during the resolution process.

- Step: Clear BGP routing information, Command: clear bgp process, Description: This command clears the BGP routing information, rolling back any changes made during the resolution process.